

Bits 23:16 **HSICAL[7:0]**: HSI16 clock calibration

These bits are initialized at startup with the factory-programmed HSI16 calibration trim value. When HSITRIM is written, HSICAL is updated with the sum of HSITRIM and the factory trim value.

Bits 15:8 **MSITRIM[7:0]**: MSI clock trimming

These bits provide an additional user-programmable trimming value that is added to the MSICAL[7:0] bits. It can be programmed to adjust to variations in voltage and temperature that influence the frequency of the MSI.

Bits 7:0 **MSICAL[7:0]**: MSI clock calibration

These bits are initialized at startup with the factory-programmed MSI calibration trim value. When MSITRIM is written, MSICAL is updated with the sum of MSITRIM and the factory trim value.

9.8.3 RCC clock configuration register (RCC_CFGR)

Address offset: 0x008

Reset value: 0x0000 0000

Access: $0 \leq \text{wait state} \leq 2$, word, half-word and byte access

1 or 2 wait states are inserted only if the access occurs during clock source switch.

From 0 to 15 wait states are inserted if the access occurs when the APB or AHB prescalers values update is on going.

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Res.	MCOPRE[2:0]			MCOSEL[3:0]				Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.
	rw	rw	rw	rw	rw	rw	rw								
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
STOPWUCK	Res.	PPRE2[2:0]			PPRE1[2:0]			HPRE[3:0]				SWS[1:0]		SW[1:0]	
	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	r	r	rw	rw

Bit 31 Reserved, must be kept at reset value.

Bits 30:28 **MCOPRE[2:0]**: microcontroller clock output prescaler

These bits are set and cleared by software.

It is highly recommended to change this prescaler before MCO output is enabled.

000: MCO divided by 1

001: MCO divided by 2

010: MCO divided by 4

011: MCO divided by 8

100: MCO divided by 16

Others: not allowed